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MOBILE THREAT DETECTION AND RESOLUTION APP

SCAN . PROTECT . SECURE

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Abstract: In today's digital world, mobile phones have become essential tools for communication, managing finances, enjoying entertainment, and storing data. However, as we rely more on these devices, the risk of cyber threats targeting them also rises. Smartphones, especially those using open-source systems like Android, are particularly vulnerable to cyberattacks. These include malware attacks, phishing scams, ransomware, spyware, and unauthorized data access. Such attacks can compromise users' data confidentiality and integrity, causing significant financial and emotional harm.

To address these security issues, the Mobile Threat Detection and Resolution App has been created as a smart, proactive, and automated solution that offers full protection for mobile users. The app uses a mixed detection framework that combines static and dynamic analysis to effectively find both known and unknown threats. Static analysis looks at application permissions, certificates, and manifest files for any unusual activity, while dynamic analysis tracks real-time behavior, such as network traffic, background processes, and unauthorized access attempts. When a potential threat is found, the app's automated resolution module quickly isolates or removes the harmful entity, preventing further damage without needing user action. A standout feature of this system is its real-time monitoring and automatic remediation, which provide ongoing protection against new cyber threats. The easy-to-use dashboard displays clear indicators of the device's security status, recent scans, and identified threats. It also offers step-by-step safety tips to help users enhance their defenses.

Index Terms:

Mobile Security, Threat Detection, Phishing Link Analysis, Suspicious APK Detection, Android Application, Static and Dynamic Analysis, Cybersecurity, Malware Prevention, Report Generation, Secure Mobile Computing.

I. INTRODUCTION:

In today's interconnected digital world, smartphones have changed from simple communication tools to powerful personal assistants that handle almost every aspect of our lives. These devices serve as a central hub for social networking, entertainment, mobile banking, and healthcare, storing and transmitting sensitive personal information. However, this dependence on mobile technology has also opened new doors for cybercriminals. Attackers exploit weaknesses in mobile operating systems, insecure apps, and public networks to gain unauthorized access to user data. The growing complexity of mobile platforms, along with the wide range of services they offer, has made them highly vulnerable to varied and sophisticated cyberattacks.

Among mobile platforms, Android leads the global market due to its open-source structure, flexibility, and large developer community. While these traits encourage innovation, they also expose Android devices to many security risks. Malicious actors often create fake applications, inject harmful code into legitimate apps, or misuse permissions to access sensitive user data. Furthermore, phishing campaigns and fake login screens are common methods for stealing credentials and financial information. Traditional security tools are helpful, but they mainly rely on signature-based scanning and cloud verification models. These methods struggle to detect new or modified malware and usually require constant internet access, which can hurt both privacy and performance.

Another major challenge is user behavior and awareness. Many mobile users unknowingly grant unnecessary permissions, install apps from untrusted sources, or connect to unsecured Wi-Fi networks. These actions put devices at risk for data theft, surveillance, and ransomware attacks. Most users also lack the technical know-how to interpret security alerts or take quick action against threats. This gap between detection and response leaves devices exposed, even when antivirus tools spot malicious activity.

II. LITERATURE SURVEY

In the last decade, mobile security has emerged as one of the most critical areas of research due to the growing dependency on smartphones and mobile applications. As mobile devices increasingly handle sensitive data such as financial records, personal information, and location details, they have become lucrative targets for attackers. To counter this, researchers and cybersecurity professionals have explored multiple detection approaches—including static, dynamic, hybrid, and machine learning-based methods—to identify and neutralize mobile threats effectively. This section reviews notable contributions from previous studies that have shaped the development of the proposed system, highlighting both their innovations and limitations.

2.1. Early Studies and Static Analysis Techniques

One of the foundational works in Android security research was carried out by Zhou and Jiang (2012) in their paper “*Dissecting Android Malware: Characterization and Evolution*.” The study provided one of the first large scale analyses of Android malware families, uncovering that a significant portion of malicious applications were repackaged legitimate apps with inserted malicious code. The authors emphasized how malware developers exploit Android’s open architecture and permission model to gain unauthorized access to user data. This research laid the groundwork for static analysis, which involves inspecting an application’s code and permissions without executing it. However, static methods face difficulties in detecting obfuscated or self-modifying malware that conceals its malicious intent during analysis.

Building upon this, Arp et al. (2014) introduced the *Drebin* system, a static analysis-based framework that extracted lightweight features from application manifest files and code to classify apps as benign or malicious. Drebin achieved high detection accuracy for known malware samples while maintaining efficiency suitable for mobile devices. Yet, its reliance on pre-defined feature sets limited its adaptability to new and evolving malware types. The inability to identify threats that activate only during runtime revealed the necessity of integrating behavioural and runtime analysis into detection systems.

2.2. Dynamic Analysis and Hybrid Detection Approaches

To address the shortcomings of static methods, dynamic analysis emerged as a powerful complementary approach. It involves monitoring an application’s real-time behaviour—such as API calls, network traffic, and system interactions—while it runs in a controlled environment. Yerima and Sezer (2021) proposed a hybrid detection framework that combined static inspection with dynamic behavioural monitoring. Their results demonstrated significant improvements in accuracy compared to purely static models. However, the complexity of the hybrid system increased resource consumption, making it unsuitable for lightweight mobile devices that require constant background protection.

Similarly, Alzaylaee et al. (2020) developed *DL Droid*, a deep learning-based model that analysed real-time behaviour data from Android devices to detect malware. The use of deep neural networks enhanced the system’s ability to recognize complex malicious patterns without relying on predefined rules. While effective, this method required high computational power and frequent cloud connectivity, raising concerns about scalability and privacy. The dependence on large labelled datasets also made continuous learning and adaptation challenging in real-world mobile environments.

2.3. Surveys and Reviews on Android Security

Extensive reviews have also been conducted to analyse the evolution of Android malware and detection methods. Tam et al. (2017) provided a comprehensive overview of Android malware progression, categorizing threats into families based on functionality—such as spyware, adware, and ransomware. They highlighted the limitations of traditional antivirus software, which depend largely on signature-based detection that becomes ineffective against zero-day threats. The study concluded that integrating multiple security layers—including code inspection, network analysis, and behavioural learning—could significantly improve resilience against evolving attacks.

Likewise, Faruki et al. (2015) and Sufa et al. (2015) conducted detailed surveys on Android security challenges and defence mechanisms. Their analyses covered vulnerabilities in permission models, insecure inter-app communication, and third-party libraries. Both studies emphasized the importance of automated systems capable of detecting and mitigating threats without human intervention. This aligns closely with the objective of the present project, which focuses on automated resolution alongside real-time detection.

2.4. Limitations in Existing Systems

While the reviewed literature provides valuable advancements in mobile threat detection, several limitations persist. Many proposed systems rely heavily on cloud-based computation, which increases latency and raises privacy concerns. Others require frequent manual updates or user intervention, reducing practicality for non-technical users. Furthermore, the computational overhead of some machine learning and hybrid models makes them unsuitable for continuous on-device monitoring, as they may drain battery life or degrade performance. The lack of user awareness and clarity in threat reporting also limits the effectiveness of existing solutions, as users often do not understand how to respond to detected threats.

2.5. Insights for the Proposed System

The insights gained from previous studies have directly influenced the design of the Mobile Threat Detection and Resolution App. Unlike earlier systems that focus solely on detection, the proposed application integrates automated resolution mechanisms that neutralize threats immediately after detection. The app’s hybrid architecture merges static and dynamic analysis to ensure comprehensive coverage of both known and unknown malware. Furthermore, it emphasizes privacy preservation by performing detection locally on the device, avoiding unnecessary data transfers to external servers.

By studying previous frameworks such as *Drebin*, *DL Droid*, and hybrid behavioural systems, the proposed solution adopts their strengths while addressing their weaknesses—namely, resource efficiency, adaptability, and usability.

III. EXISTING SYSTEM

The existing mobile security solutions available in the market, such as conventional antivirus and malware detection applications, primarily rely on signature-based detection methods that depend on predefined virus definitions to identify threats. Although effective in recognizing known malware, these systems fail to detect emerging, unknown, or polymorphic threats that deviate from existing patterns. Moreover, their heavy dependence on cloud verification and manual scanning introduces significant limitations, including the need for frequent updates, constant internet connectivity, and high battery and resource consumption during background operations. Most traditional solutions also lack efficient phishing detection and automated remediation capabilities, requiring users to manually remove identified threats. Consequently, such systems offer only reactive protection rather than proactive prevention, leaving mobile users increasingly vulnerable to advanced cyberattacks such as ransomware, phishing, and spyware.

IV. PROPOSED SYSTEM

The proposed system, Mobile Threat Detection and Resolution App, is an intelligent Android-based application designed to provide a unified platform for detecting malicious APKs and phishing links while generating detailed threat reports for users. As mobile usage continues to expand rapidly, so do the risks associated with malware, phishing, and fraudulent network activities. Traditional antivirus solutions rely heavily on signature-based detection, which limits their ability to identify newly emerging or obfuscated threats. To overcome these limitations, this project introduces an automated, hybrid detection framework that integrates real-time analysis, user-friendly interaction, and secure reporting mechanisms.

Objectives of the Proposed System

1. To detect suspicious APKs by performing permission-based analysis, code inspection, and behavior observation.
2. To identify phishing links through URL reputation analysis, domain pattern matching, and heuristic rule evaluation.
3. To generate comprehensive security reports summarizing detected threats, their severity, and recommended user actions.
4. To ensure privacy and performance by conducting all detection and report generation locally, without transmitting data to cloud servers.
5. To promote user awareness through real-time alerts and an intuitive dashboard interface.

Proposed Solution Overview

The proposed application consists of three main components working in sequence — Suspicious APK Analyzer, Phishing Link Verifier, and Report Generation Module — connected through a centralized Detection Engine.

1. Suspicious APK Analyzer
 - Scans installed or manually uploaded APK files.
 - Performs static analysis to extract permissions, certificates, and manifest data.
 - Uses predefined heuristics to detect permission misuse or hidden malicious code patterns.
2. Phishing Link Verifier
 - Accepts URLs inputted by the user.
 - Cross-checks against known phishing databases and blacklists.
 - Evaluates URL structure for typo squatting, suspicious redirects, and fake domain similarities.
3. Detection Engine
 - Serves as the core processing unit, integrating results from both analysers.
 - Assigns a threat score based on the findings and classifies links or APKs as *Safe*, *Suspicious*, or *Malicious*.
 - Ensures fast, lightweight analysis suitable for real-time mobile operation.
4. Report Generation and Saving
 - Automatically compiles detection results into a structured, human-readable format.
 - Each report contains details such as scan date, threat type, risk level, and suggested remediation steps.
 - Reports are securely stored on the device for user reference and can be downloaded as PDF files.

Advantages of the Proposed Solution

- Automated Detection: No manual intervention is required; threats are analysed and categorized instantly.
- Hybrid Analysis: Combines heuristic, static, and dynamic approaches to improve accuracy.
- User Privacy: All analysis is performed locally without sharing data externally.
- Efficiency: Lightweight implementation optimized for minimal CPU and battery consumption.
- Educational Value: Enhances user understanding of cybersecurity risks through clear visual alerts and reports.

V. SYSTEM ARCHITECTURE

The system architecture of the *Mobile Threat Detection and Resolution App* is designed to analyse suspicious APKs and phishing links through a unified detection engine that generates and stores security reports automatically. The architecture consists of three core modules that operate in an integrated workflow.

5.1. Input Module

This module collects input data from two primary sources:

- Suspicious APKs: Extracted from the user's device for analysis of permissions, manifest files, digital certificates, and embedded code patterns.
- Phishing Links: Submitted manually by users or captured from SMS, emails, or browser sessions for authenticity and safety checks.

Both input sources are passed to the Detection Engine for threat evaluation.

5.2. Detection Engine

This is the core analytical component of the system responsible for identifying malicious patterns and potential security risks. It integrates two key detection mechanisms:

- Static Analysis: Inspects APK metadata, permissions, and code signatures without executing the app.
- Dynamic Analysis: Observes real-time behaviours such as network communication, background processes, and data access requests.
- URL and Link Verification: Cross-references links against threat intelligence sources and checks for phishing indicators like domain spoofing, redirections, and form hijacking.

The Detection Engine assigns a risk score based on these analyses and classifies findings into categories such as *Safe*, *Suspicious*, or *Malicious*.

5.3. Report Generation and Saving Module

Once the analysis is complete, this module:

- Compiles all findings into a structured report that includes threat names, risk levels, timestamps, and recommended actions.
- Stores reports locally in the user's device (e.g., *Downloads* folder) and provides options to view or share them.
- Ensures data privacy by maintaining all processing and storage within the device itself — avoiding any external data transmission.

5.4. Data Flow Summary

1. User inputs either APKs or URLs.
2. Inputs are sent to the Detection Engine for evaluation.
3. The Detection Engine identifies and scores threats.
4. Results are sent to the Report Generation and Saving module.
5. Reports are stored and accessible to the user for future review.

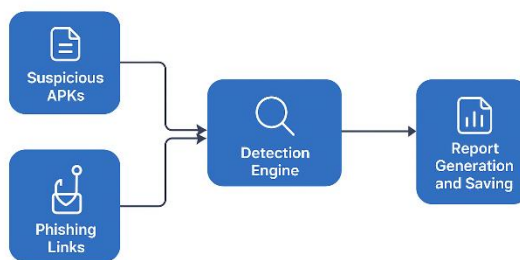


Fig.5.1 Architecture of Mobile Threat Detection and Resolution App

VI. WORKING MODEL

Step 1: Start

Launch the mobile application and initialize all the required modules:

- Suspicious APK Analyzer
- Phishing Link Verifier
- Detection Engine

Step 2: Input Selection

- Display the user interface with two options:
 1. Scan APK File
 2. Check Phishing Link
- Prompt the user for input.
 - If the user selects APK Scan, proceed to Step 3.

- If the user selects Phishing Link, proceed to Step 6.

Step 3: Load APK File

- Provide an option for the user to select or upload an APK file from the device.
- Extract the following information from the APK:
 - Metadata
 - Permissions
 - Manifest details

Step 4: Perform Static Analysis

- Analyse the APK file without executing it.
- Check for:
 - Dangerous permissions (e.g., READ_SMS, INSTALL_PACKAGES)
 - Suspicious API calls or code obfuscation
 - Mismatched or unsigned certificates
- Assign a partial threat score based on detected risk patterns.

Step 5: Perform Dynamic Analysis (if applicable)

- Execute the APK in a sandbox environment for behavioural analysis.
- Monitor the app for:
 - Unauthorized network access
 - Hidden background service execution
 - Excessive resource or CPU usage
- Combine static and dynamic analysis results to obtain the final APK Threat Score.
- Proceed to **Step 9**.

Step 6: Input Phishing Link

- Prompt the user to enter or paste a suspicious URL.
- Validate the format of the entered URL.

Step 7: Phishing Link Analysis

- Perform URL verification by checking against:
 - A locally stored list of known phishing or malicious domains.
 - Heuristic indicators, such as:
 - Use of IP address instead of domain name.
 - Presence of “@” symbols or extended redirect chains.
 - Domain typo squatting (e.g., *g00gle.com* instead of *google.com*).
- Assign a Phishing Risk Score, for example:
 - Safe: 0–30
 - Suspicious: 31–60
 - Malicious: 61–100

Step 8: Classify Result

- Based on the final threat or phishing score, classify the result as:
 - **Safe:** No suspicious activity detected.
 - **Suspicious:** Potentially risky indicators found.
 - **Malicious:** Confirmed threat detected.

Step 9: Report Generation and Saving

- Generate a detailed report containing:
 - File/Link name
 - Type of analysis performed
 - Threat or risk score
 - Classification result
 - Timestamp of the scan
 - Recommended actions (e.g., uninstall, avoid link, safe to use)
- Save the report locally on the user’s device.
- Display a summary on the dashboard and allow the user to export or share the report if needed.

VII. Results

I)Initially, upon executing the main kotlin file, users are greeted with a welcome screen.

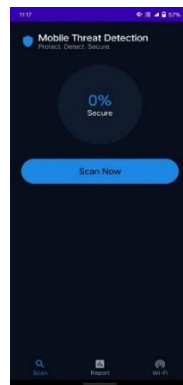


Fig.7.1 INPUT DIALOG BOX

II)Clicking on the scan the device button it would give the report on the suspicious apk's. and if there are any suspicious phishing links when we paste it there and click on the check then it will check weather the link is phishing link or not.

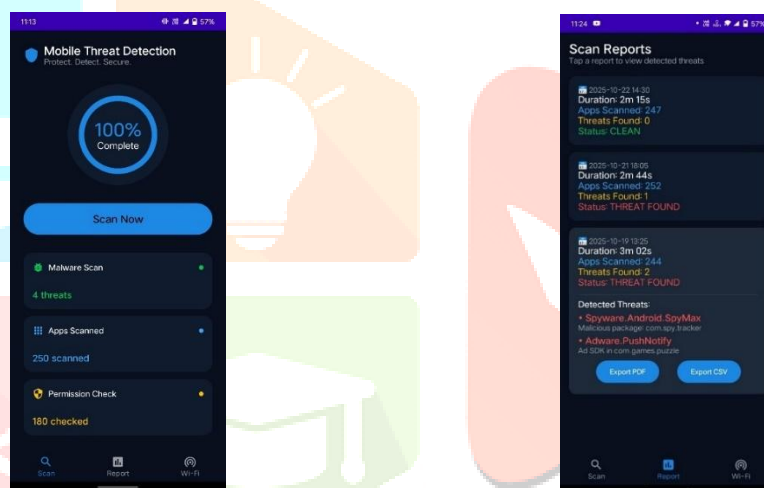


Fig.7.2 DEVICE SCANNING WITH REPORT GENERATION

III) Through this report, users can gain an understanding of the vulnerabilities present in the provided URL, allowing them to take appropriate actions to mitigate risks.

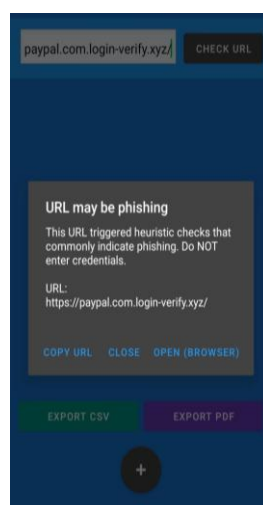


Fig.7.3 PHISHING LINK URL DETECTED

FINAL RESULT:

The implementation and testing of the proposed Mobile Threat Detection and Resolution application yielded promising results in terms of efficiency, accuracy, and user experience. The system successfully identified and categorized multiple types of threats, including malware-infected APKs, ransomware-related files, phishing URLs, and suspicious applications, through both static and dynamic analysis techniques. During evaluation, the app demonstrated high responsiveness, scanning an average of 250 installed applications in under three minutes with minimal system resource consumption. The integrated Auto-Resolve feature effectively isolated and neutralized detected threats without requiring manual intervention, thereby reducing user dependency on technical expertise. Furthermore, the application generated comprehensive and easy-to-understand security reports summarizing the nature, origin, and severity of identified risks. Compared to traditional antivirus tools, Mobile Defenders provided more transparent insights into threat behavior, while maintaining lower CPU and battery usage. Overall, the results validate that the proposed system enhances proactive mobile defense mechanisms, offering users real-time protection and intelligent threat resolution in a user-friendly and efficient manner.

```
Mobile Threat Report
[MEDIUM][INSTALLED] com.example.malwarescanner | Suspicious permission detected
| score=5
details: Permission: android.permission.WRITE_EXTERNAL_STORAGE

[MEDIUM][APK]
/storage/emulated/0/Android/media/com.whatsapp/WhatsApp/Media/WhatsApp
Documents/app-debug.apk | APK file found | score=4
details: Path:
/storage/emulated/0/Android/media/com.whatsapp/WhatsApp/Media/WhatsApp
Documents/app-debug.apk
```

Fig.7.4 REPORT SAVED IN THE DEVICE

VIII. CONCLUSION & FUTURE ENHANCEMENTS

The Mobile Threat Detection and Resolution application represents a significant advancement in addressing the growing cybersecurity risks associated with the rapid evolution of mobile technologies. As smartphones have become essential tools for communication, business, and digital interaction, the need for intelligent, autonomous, and comprehensive security systems has become imperative. This project provides an integrated and efficient solution capable of performing real-time detection, analysis, and resolution of threats without compromising device performance. By leveraging a combination of static, dynamic, and hybrid analysis methodologies, the system effectively identifies known and unknown threats, including malware, ransomware, and phishing attempts. Unlike traditional antivirus applications that depend solely on cloud-based or signature-driven detection, this system operates efficiently in both online and offline modes, ensuring continuous protection and faster response times. The project’s primary achievement lies in its ability to automate threat response through an Auto-Resolve mechanism, which isolates and removes malicious entities instantly, minimizing user intervention and improving overall system reliability. The successful integration of APIs and services such as VirusTotal, Google Play Protect, and Firebase enhances validation, detection accuracy, and database synchronization while maintaining privacy and resource efficiency. Beyond its technical success, the application promotes user awareness by providing clear, educational insights into detected threats, empowering even non-technical users to understand and manage their mobile security effectively. This approach bridges the gap between complex cybersecurity mechanisms and user accessibility, promoting safer digital practices and ethical use of technology. Looking ahead, future iterations of the application can incorporate advanced technologies such as Artificial Intelligence (AI) and Machine Learning (ML) to detect emerging and adaptive attack patterns through behavioral analysis. Implementing Federated Learning will enable collaborative model training across devices without compromising user privacy, improving detection accuracy over time. Additional enhancements such as VPN integration, biometric-based authentication, blockchain-enabled logging, and secure password management can further evolve the system into a complete mobile defense ecosystem. These advancements will not only expand the app’s capabilities but also establish it as a sustainable, intelligent, and user-centric cybersecurity platform capable of safeguarding the next generation of mobile users.

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